Numerical Integration

The need often arises for evaluating the definite integral of a function that has no explicit antiderivative or whose antiderivative is not easy to obtain. The basic method involved in approximating $\int_a^b f(x)dx$ is called numerical quadrature. It uses a sum $\sum_{i=0}^{n} a_i f(x_i)$ to approximate $\int_a^b f(x)dx$.

The methods of quadrature in this section are based on the interpolation polynomials. The basic idea is to select a set of distinct nodes $\{x_0, \dots, x_n\}$ from the interval $[a, b]$. Then integrate the Lagrange interpolating polynomial,

$$P_n(x) = \sum_{i=0}^{n} f(x_i)L_i(x)$$

and its truncation error term over $[a, b]$ to obtain,

$$\int_a^b f(x)\,dx = \int_a^b \sum_{i=0}^{n} f(x_i)L_i(x)\,dx + \int_a^b \prod_{i=0}^{n}(x - x_i)\frac{f^{(n+1)}(\xi(x))}{(n+1)!}dx$$

$$= \sum_{i=0}^{n} a_i f(x_i) + \frac{1}{(n+1)!}\int_a^b \prod_{i=0}^{n}(x - x_i) f^{(n+1)}(\xi(x))\,dx,$$

Where $\xi(x)$ is in $[a, b]$ for each x and

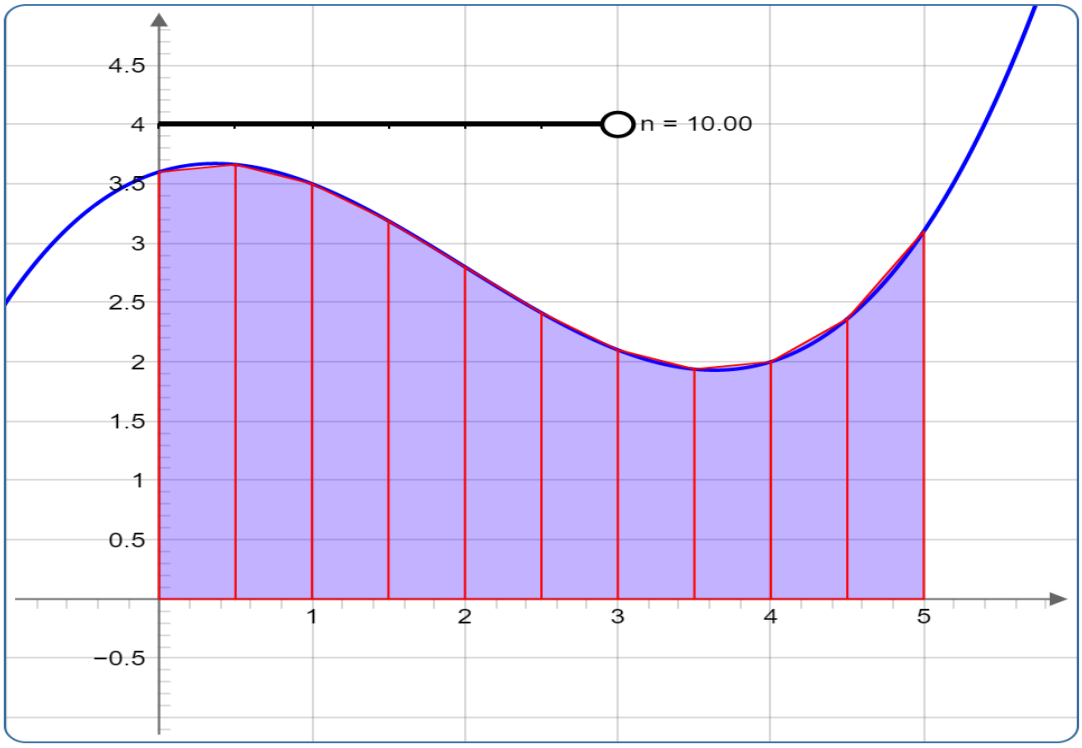
$$a_i = \int_a^b L_i(x)dx\,, \quad \text{for each } i = 0,1,2 \dots, n.$$

The quadrature formula is, therefore,

$$\int_a^b f(x)dx \approx \sum_{i=0}^{n} a_i f(x_i)\,,$$

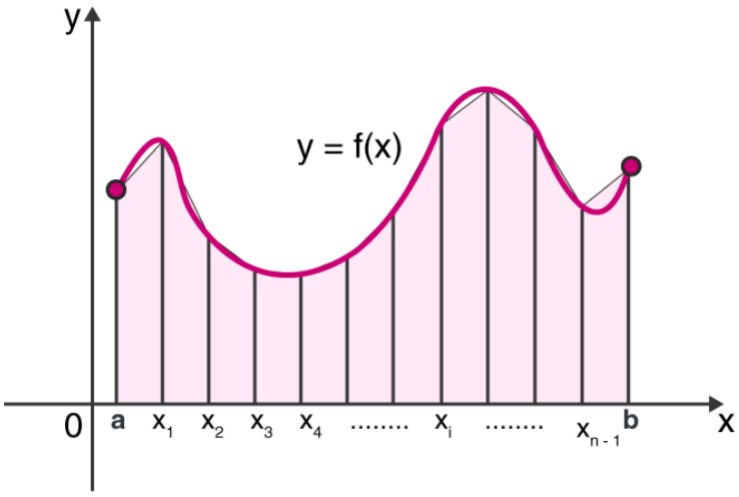
With error given by

$$E(f) = \frac{1}{(n+1)!}\int_a^b \prod_{i=0}^{n}(x - x_i) f^{(n+1)}(\xi(x))\,dx.$$



The Trapezoidal Rule

This technique is a much more accurate way to approximate area beneath a curve. To construct the trapezoids, you mark the height of the function at the beginning and end of the width interval, then connect the two points. However, this method requires you to memorize a formula.



Definition

Let f be continuous on $[a, b]$. The Trapezoidal Rule for approximating $\int_a^b f(x)\,dx$ is given by

$$\frac{b-a}{2n}[f(a)+2f(x_1)+2f(x_2)+\cdots+2f(x_{n-1})+f(b)]$$

The Trapezoidal Rule A Second Glimpse

$$\int_a^b f(x)\,dx=\frac{h}{2}[y_0+2y_1+2y_2+\cdots+2y_{n-1}+y_n]$$

where $[a, b]$ is partitioned into n subintervals of equal length.

$$\Delta x=h=\frac{b-a}{n}$$

Ex 1: Approximate the area beneath $f(x)=x^2+1$ on the interval $[0,3]$ using the Trapezoidal Rule with $n=5$ trapezoids.

Solution:

$$\Delta x=h=\frac{b-a}{n}=\frac{3-0}{5}=\frac{3}{5}$$

x	0	$\frac{3}{5}$	$\frac{6}{5}$	$\frac{9}{5}$	$\frac{12}{5}$	3
$f(x)=y$	1	$\frac{34}{25}$	$\frac{61}{25}$	$\frac{106}{25}$	$\frac{169}{25}$	10

$$I=\frac{b-a}{2n}[y_0+2y_1+2y_2+2y_3+2y_4+y_5]$$

$$I=\frac{3}{2(5)}\left[1+2\left(\frac{34}{25}\right)+2\left(\frac{61}{25}\right)+2\left(\frac{106}{25}\right)+2\left(\frac{169}{25}\right)+10\right]$$

$$I=\frac{3}{10}\left[1+\frac{68}{25}+\frac{122}{25}+\frac{212}{25}+\frac{338}{25}+10\right]$$

$$I=\frac{3}{10}\left[\frac{25+68+122+212+338+250}{25}\right]$$

$$I=\frac{3}{10}\left[\frac{1015}{25}\right]=\frac{609}{50}=12.18$$

Ex 2: Approximate the area beneath $y=sinx$ on the interval $[0,\pi]$ using the Trapezoidal Rule with $n=4$ trapezoids.

Solution:

$$\Delta x=h=\frac{b-a}{n}=\frac{\pi-0}{4}=\frac{\pi}{4}$$

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
$f(x)=y$	0	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{2}}{2}$	0

$$I=\frac{b-a}{2n}[y_0+2y_1+2y_2+2y_3+y_4]$$

$$I=\frac{\pi}{2(4)}\left[0+2\left(\frac{\sqrt{2}}{2}\right)+2(1)+2\left(\frac{\sqrt{2}}{2}\right)+0\right]$$

$$I=\frac{\pi}{8}[0+\sqrt{2}+2+\sqrt{2}+0]$$

$$I=\frac{\pi}{8}[2+2\sqrt{2}]$$

$$I\approx 1.894$$

Ex 3: Given the following experiment data determine the approximate area enclosed by a curve passing through the data and the x-axis by the trapezoidal rule.

x	1	2	3	5
$f(x)=y$	2	4	5	7

Solution:

$$I=I_1+I_2$$

$$I_1=\frac{h_1}{2}[f(1)+2f(2)+f(3)],h_1=2-1=1$$

$$I_2=\frac{h_2}{2}[f(3)+f(5)],h_2=5-3=2$$

$$I=\frac{1}{2}[2+2(4)+5]+\frac{2}{2}[5+7]=19.5$$

Ex 4: Approximate the area under the curve $y=f(x)$ between $x=0$ and $x=8$ using Trapezoidal Rule with $n=4$ subintervals. A function $f(x)$ is given in the table of values.

x	0	2	4	6	8
$f(x)$	3	7	11	9	3

Solution:

The Trapezoidal Rule formula for $n=4$ subintervals is given as:

$$I = \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)]$$

Here the subinterval width $\Delta x = 2$.

Now, substitute the values from the table, to find the approximate value of the area under the curve.

$$A \approx I = \frac{2}{2}[3 + 2(7) + 2(11) + 2(9) + 3]$$

$$A \approx I = 3 + 14 + 22 + 18 + 3 = 60$$

Therefore, the approximate value of area under the curve using Trapezoidal Rule is 60.

Ex 5: Approximate the area under the curve $y = f(x)$ between $x = -4$ and $x = 2$ using Trapezoidal Rule with $n = 6$ subintervals. A function $f(x)$ is given in the table of values.

x	-4	-3	-2	-1	0	1	2
$f(x)$	0	4	5	3	10	11	2

Solution:

The Trapezoidal Rule formula for $n = 6$ subintervals is given as:

$$I = \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + 2f(x_5) + f(x_6)]$$

Here the subinterval width $\Delta x = 1$.

Now, substitute the values from the table, to find the approximate value of the area under the curve.

$$A \approx I = \frac{1}{2}[0 + 2(4) + 2(5) + 2(3) + 2(10) + 2(11) + 2]$$

$$A \approx I = \frac{1}{2}[8 + 10 + 6 + 20 + 22 + 2] = \frac{68}{2} = 34$$

Therefore, the approximate value of area under the curve using Trapezoidal Rule is 34.